HYEON LEE

hyeonlee@unc.edu | https://hyeonlee-stats.github.io

EDUCATION

University of North Carolina at Chapel Hill Aug. 2022 - Present (Expected May 2027)
Ph.D. Candidate in Statistics and Operations Research, advised by Professor J. S. Marron

Seoul National University Mar. 2020 - Feb. 2022
M.S. in Statistics, advised by Professor Byeong U. Park

Korea Advanced Institute of Science and Technology Mar. 2015 - Feb. 2020
B.S. in Mathematical Science, minor in Industrial and Systems Engineering

HONORS AND AWARDS

Outstanding Achievement Award Dec. 2023
Department of Statistics and Operations Research, UNC Chapel Hill

Presidential Science Scholarship 2015 - 2019
Korea Student Aid Foundation, Republic of Korea

RESEARCH INTERESTS

Principal Subsimplex Analysis
Developing statistical methodologies for dimension reduction and identification of mode of variation for compositional data via nested sequence of subsimplices.

Functional and Shape Analysis of Biomechanical Curves
Analyzing regression models for gait curves with repeated measurements using Elastic Shape Analysis.

Functional Analysis using Wasserstein-Fisher-Rao Distance
Developing a workflow for analyzing functional data using the Wasserstein-Fisher-Rao distance for generalization of optimal transport framework.

Data Integration for Numerical and Categorical Variables
Identifying shared and individual modes of variation using low-rank models for categorical and numerical variables.

PUBLICATIONS & PREPRINTS

Peer-Reviewed Publications

[1] **Lee**, **H.**, Hingee, K. L., Scealy, J. L., Wood, A. T., Grunsky, E., and Marron, J. S. "Principal subsimplex analysis". *Journal of computational and graphical statistics* (2026). accepted.

[2] Buck, A. N., Borgert, J. E., **Lee**, **H.**, Arbeeva, L., Golightly, Y. M., Loeser, R. F., Callahan, L. F., Messier, S. P., Pietrosimone, B., Nelson, A. E., and Marron, J. S. "Unsupervised machine learning identifies shared biomechanical and clinical variation in individuals with knee osteoarthritis and high bmi: a secondary analysis of the idea trial". *Arthritis care & research* (2026). accepted.

Peer-Reviewed Abstracts

[3] Buck, A. N., Borgert, J. E., **Lee**, **H.**, Arbeeva, L., Golightly, Y. M., Loeser, R. F., Messier, S. P., Pietrosimone, B., Nelson, A. E., and Marron, J. S. "Vertical ground reaction force variability is associated with clinical features in individuals with knee oa and overweight/obesity: a novel machine learning analysis of the IDEA trial". *Osteoarthritis and cartilage* 33 (2025), S160–S161.

Submitted Papers & Preprints

[4] **Lee**, **H.**, Armitano-Lago, C., Kelly, D. K., Arbeeva, L., Borgert, J., Buck, A. N., Messier, S. P., Nelson, A. E., and Marron, J. S. "Disentangling the effects of diet, exercise, walking speed, and weight loss on vertical ground reaction forces in knee osteoarthritis: findings from the IDEA trial". Submitted at *Journal of Applied Biomechanics*.

[5] **Lee**, **H.**, Marron, J. S., Grunsky, E., and Scealy, J. L. "Dimension reduction of compositional data through mixtures of latent source compositions". Submitted at *Journal of the Royal Statistical Society Series B: Statistical Methodology*.

SOFTWARE

- R package **psacomp**, available at GitHub. (`https://github.com/hyeon-lee-stat/psacomp`)

PRESENTATIONS

Principal Subsimplex Analysis — Contributed Paper
Joint Statistical Meetings — *August 2025*

Principal Subsimplex Analysis — Poster Presentation
Object Oriented Data Analysis and its Applications — *May 2025*

WORKSHOPS & SCHOOLS

Introductory School: Geometric Statistics
Hausdorff Research Institute for Mathematics — *September 2026*

PROFESSIONAL SERVICE

Referee for *Proceedings of the American Mathematical Society* — *1 manuscript*

TEACHING

Teaching at UNC Chapel Hill

- Instructor for STOR 155: Introduction to Data Models and Inference — Fall 2025
- Teaching assistant for STOR 155: Introduction to Data Models and Inference — Fall 2023
- Teaching assistant for STOR 535: Probability for Data Science — Spring 2023
- Teaching assistant for STOR 435: Introduction to Probability — Fall 2022

Teaching at Seoul National University

- Teaching assistant for 033.020 Statistics Lab. — Fall 2020, Spring 2021, Fall 2021